

Lab 08 Template

Some questions require multiple parts to be answered, be sure to discuss them in full.

Part 01:

1. **Screenshot of the output from** `openssl rsa -text -in <netid>_private_key.pem`
(10 points)

Private-Key: (2048 bit, 2 primes)

modulus:

00:a2:44:02:81:0a:ed:ae:8d:e0:d0:6a:3a:63:4f:
dd:dd:45:04:8b:ee:bf:8d:74:6d:80:a5:75:a1:13:
da:c4:d9:9f:89:66:25:e8:fa:9c:ea:39:17:06:84:
3b:39:1d:41:21:b7:7c:ee:1b:2f:2c:35:d4:43:29:
71:af:d7:37:58:56:79:de:25:fa:d5:39:14:43:77:
49:df:05:00:fe:90:a1:99:40:cb:d3:89:a3:be:94:
28:22:24:77:07:1a:1d:0d:9f:0a:42:6d:d7:b0:fd:
d0:68:12:7b:e1:55:28:76:0d:71:74:dc:ec:62:54:
ac:ec:fa:92:76:ca:88:c4:f0:5a:13:52:89:54:3f:
30:3a:bd:94:a3:c7:da:d0:8f:ca:73:2e:4e:6b:7d:
ef:cf:ae:31:56:6a:cc:9e:d5:0b:36:87:f9:ab:b8:
35:cf:50:4c:c0:b0:13:86:0f:fa:7f:9d:3a:b3:7b:
9b:63:dc:54:4a:45:6e:92:a0:ce:e1:47:59:eb:20:
51:e0:79:2d:ec:1b:1a:d8:7f:ab:42:40:73:8e:76:
36:a9:d3:79:e7:61:67:5b:47:b3:02:4f:25:f9:d7:
38:33:41:31:00:16:0b:c6:5c:44:6a:24:93:bb:db:
c2:23:de:3a:6a:e5:62:41:68:3a:39:19:45:54:c5:
71:09

publicExponent: 65537 (0x10001)

privateExponent:

08:31:e8:1b:22:9f:fe:09:fd:f9:cd:f6:49:f0:0d:
7e:9b:24:3a:24:4c:cf:4a:3b:2c:dd:43:b3:66:b5:
cc:98:48:65:8a:34:f8:44:63:ea:b3:b1:bf:d6:00:
c9:b1:cf:b4:1c:b8:b6:01:1b:20:f5:4d:00:36:ab:
c9:06:2a:58:e4:9b:d3:5b:81:5a:05:bf:ca:12:7b:
0c:1a:36:c2:79:da:77:76:1d:75:29:af:e2:89:9d:
51:f0:1b:da:23:78:51:28:f5:93:2f:9f:4e:39:db:
58:78:df:d7:1c:53:bb:f6:92:2d:7a:94:d8:f8:79:
77:f4:70:5e:33:27:5f:79:71:66:a0:c6:32:53:3f:

:
█

```
publicExponent: 65537 (0x10001)
privateExponent:
  08:31:e8:1b:22:9f:fe:09:fd:f9:cd:f6:49:f0:0d:
  7e:9b:24:3a:24:4c:cf:4a:3b:2c:dd:43:b3:66:b5:
  cc:98:48:65:8a:34:f8:44:63:ea:b3:b1:bf:d6:00:
  c9:b1:cf:b4:1c:b8:b6:01:1b:20:f5:4d:00:36:ab:
  c9:06:2a:58:e4:9b:d3:5b:81:5a:05:bf:ca:12:7b:
  0c:1a:36:c2:79:da:77:76:1d:75:29:af:e2:89:9d:
  51:f0:1b:da:23:78:51:28:f5:93:2f:9f:4e:39:db:
  58:78:df:d7:1c:53:bb:f6:92:2d:7a:94:d8:f8:79:
  77:f4:70:5e:33:27:5f:79:71:66:a0:c6:32:53:3f:
  f0:49:05:f5:d9:70:66:0e:3d:05:5b:4e:80:cb:f9:
  a7:8c:30:4b:5c:6b:ff:bf:32:ce:34:9a:26:4b:6d:
  49:e8:78:98:3a:c6:48:8d:80:62:8e:ba:96:10:c3:
  48:3e:9c:38:f4:d9:76:8d:60:c4:9d:64:99:fb:28:
  1d:58:c2:14:9e:39:43:b8:bd:72:94:21:7f:37:ba:
  c0:2b:c0:25:b4:1c:1f:ef:1c:c7:a2:b6:5f:de:bd:
  d5:ca:0a:05:6c:47:61:e1:61:09:e9:f3:30:a6:1e:
  6c:1b:f1:9b:48:51:ab:41:cb:f0:00:7a:fa:00:c1:
  2b
prime1:
  00:db:62:98:9b:ea:54:98:08:09:77:9d:15:ee:a6:
  1b:cd:5d:3b:dc:6c:7f:f3:a5:4e:be:2c:ee:0d:1e:
  fe:fb:47:b4:65:48:21:e0:59:2f:a1:22:8c:74:ef:
  59:ab:14:09:70:64:86:a3:1d:0c:4c:fc:df:90:c2:
  55:4f:9c:17:14:e4:92:be:6f:a8:b3:c3:72:e7:1c:
  3f:71:7b:c1:18:4c:7b:57:1a:83:0a:bb:02:4b:65:
  be:13:68:00:3f:c4:02:37:fe:81:42:f4:19:4c:0a:
  94:ec:77:2c:50:ec:96:1c:15:0b:57:45:9e:e7:68:
  02:b7:50:5e:03:ee:16:c9:ef
prime2:
:
```


prime2:

00:bd:58:f0:50:90:0d:40:26:90:34:c7:ef:e5:45:
06:02:b6:31:ac:4b:23:e6:9f:82:f3:5e:f4:70:14:
17:81:54:1a:b9:05:1d:f0:7f:fb:60:79:62:6c:40:
13:87:6d:31:58:29:a6:45:17:c1:9e:7b:3e:c6:4e:
eb:e2:f1:6f:db:44:02:79:d6:d1:1d:4d:6f:dd:79:
4b:98:61:f7:e4:ac:60:6e:1a:1c:36:e8:7e:79:1e:
3d:d5:95:5f:20:ca:ff:0f:8e:84:c6:80:22:ff:95:
26:78:bb:d6:1a:e1:03:05:e8:e5:72:44:b8:d0:17:
39:4e:6e:da:c3:30:12:4c:87

exponent1:

7c:56:11:f5:f3:c6:03:f0:32:25:67:93:21:91:65:
1a:6c:27:dc:12:66:26:19:5f:5a:6b:2e:73:a3:48:
91:f3:9e:aa:f4:49:b6:d3:f5:29:96:e2:11:f8:01:
52:f3:65:32:1c:d7:2d:a4:ed:ee:c8:36:90:dd:82:
ea:3e:a5:da:42:7d:64:8e:f6:07:03:d3:a0:66:23:
12:d7:37:a5:e4:da:d6:90:01:0a:d8:d8:41:11:25:
bb:eb:ea:90:39:2f:a4:ba:8a:59:fe:ec:43:6b:8f:
e1:c0:dd:0d:f4:38:53:2c:1e:24:55:21:ed:36:c6:
ab:4c:52:2c:5a:08:f2:63

exponent2:

1f:25:eb:5a:85:67:f8:9e:9f:68:5f:5f:a2:6a:a9:
63:76:cf:53:15:85:51:61:fa:2b:c8:fa:13:19:eb:
07:f6:6a:f2:f6:d9:86:ff:b6:bd:97:1a:4e:ce:77:
a7:f6:99:80:74:51:8a:da:2e:9a:c0:91:82:13:e7:
90:0c:c1:a6:e3:58:38:ec:d2:b7:ac:4e:73:b3:89:
71:06:e4:bd:b4:71:a4:8b:1d:0b:17:51:fe:e0:af:
b1:5c:02:2b:25:d9:b1:98:73:55:28:2d:19:b8:52:
7c:d8:01:8f:a8:66:53:e9:8d:1d:30:1a:ea:f8:77:

:

coefficient:

42:55:c9:6a:9e:5f:a7:8d:82:64:99:88:a2:ae:c1:
b1:c0:98:db:7c:e4:1e:e2:a4:c2:31:1b:13:62:38:
77:51:6a:81:87:13:e0:86:5e:e0:81:fc:3b:81:36:
f5:c7:27:e2:d5:3b:9a:b1:c0:6d:35:3f:97:7b:a9:
6e:7f:8f:f0:87:9d:c9:d9:9c:43:ac:1e:f9:98:43:
ef:6a:12:df:30:0f:88:e1:1f:e1:9c:b8:8d:c3:fb:
67:ec:99:1f:4a:b7:df:32:27:48:a5:ad:38:14:fb:
2e:11:89:75:e0:e7:e5:6a:8c:44:e0:7c:d8:d9:a9:
c8:3b:63:42:de:00:db:98

-----BEGIN PRIVATE KEY-----

MIIEvAIBADANBgkqhkiG9w0BAQEFAASCbKYwggSiAgEAAoIBAQCIRAKBCu2ujeDQ
ajpjT93dRQSL7r+NdG2ApXWhE9rE2Z+JZiXo+pzqORcGhDs5HUEht3zuGy8sNdRD
KXGv1zdYVnneJfrVORRDd0nfbQD+kKGZQMvTia0+lCgiJHcHGh0NnwpCbdew/dBo
EnvhVSh2DXF030xiVKzs+pJ2yojE8FoTUolUPzA6vZSjx9rQj8pzLk5rfe/PrjFW
asye1Qs2h/mruDXPUEzAsBOGD/p/nTqze5tj3FRKRW6Som7hR1nrIFHgeS3sGxrY
f6tCQH00djap03nnYWdbR7MCTyX51zgzQTEAFglGXERqJJ0728Ij3jpbq5WJBaDo5
GUVUxXEJAgMBAAECggEACDHoGyKf/gn9+c32SfANfpsk0iRMz0o7LN1Ds2a1zJhI
ZYo0+ERj6rOxv9YAybHPtBy4tgEbIPVNADaryQYqWOSb01uBWgW/yhJ7DBo2wnna
d3YddSmv4omdUfAb2iN4USj1ky+fTjnbWHjf1xxTu/aSLXqU2Ph5d/RwXjMnX3lx
ZqDGMlM/8EkF9dlwZg49BVtOgMv5p4wwS1xr/78yzjSaJkttSeh4mDrGSI2AYo66
lhDDSD6cOPTZdo1gxJ1kmfsoHVjCFJ45Q7i9cpQhfze6wCvAJbQcH+8cx6K2X969
1coKBWxHYeFhCenzMKYebBvxm0hRq0HL8AB6+gDBKwKBgQDbYpib6LSYCAL3nRXu
phvNXTvcBH/zpU6+L04NHv77R7RlSCHgWS+hIox071mrFAlwZiajHQxM/N+QwlVP
nBcU5JK+b6izw3LnHD9xe8EYTHtXGoMKuwJLZb4TaAA/xAI3/oFC9BlMCpTsdyxQ
7JYcFQtXRZ7naAK3UF4D7hbJ7wKBgQC9WPBQkA1AJpA0x+/lRQYctjGsSyPmn4Lz
XvRwFBeBVBq5BR3wf/tgeWJsQB0HbTFYKaZFF8Geez7GTuvi8W/bRAJ51tEdTW/d
eUuYYffkrGBuGhw26H55Hj3VlV8gyv8PjoTGgCL/lSZ4u9Ya4QMF60VyRLjQFzlo
btrDMBJMhwKBgHxWEfXzxpWmMiVnkyGRZRpsJ9wSZiYZX1prLn0jSJHznqr0SbbT
9SmW4hH4AVLzZTIc1y2k7e7INpDdguo+pdpCfWS09gcD06BmIxLXN6Xk2taQAQrY
2EERJbvr6pA5L6S6iln+7ENrj+HA3Q300FMshIRVie02xqtMUixaCPJjAoGAHyXr
:

2. **Comparison of the same/different values observed across the extra generated keys**

Output of key2 for reference

Private-Key: (2048 bit, 2 primes)

modulus:

00:a0:c3:ae:17:76:ca:1e:8c:2c:7d:b7:9c:c8:d1:
38:48:13:27:37:36:63:bd:9f:7f:fd:21:61:76:39:
23:89:6a:a0:e3:c1:07:f2:6a:b0:fa:b1:d4:33:2a:
a2:8a:2a:63:50:7a:0d:c3:2f:c8:55:37:d5:54:a6:
b9:5b:0e:ba:a9:84:5b:d1:95:23:d3:c0:c6:49:bc:
19:8a:09:81:a8:4d:10:bc:bc:56:f0:22:7f:23:3c:
c5:d7:bd:c0:2a:02:01:57:e5:36:f7:34:d8:65:28:
23:63:02:d6:cc:f2:b3:74:35:b5:e0:80:4b:03:22:
05:69:57:d8:e7:79:30:85:47:f6:d3:d9:18:94:08:
67:3d:2b:fd:5d:15:cb:5b:5f:dc:d9:72:0f:02:98:
b2:2a:85:5b:c3:21:f6:b3:d0:d1:91:1d:81:4b:07:
a4:87:e8:46:4a:c7:d1:d9:7a:76:74:6a:89:2b:81:
e2:e9:01:43:0b:d7:c8:a4:8c:2d:78:36:3c:f2:5f:
c2:b4:7c:23:f8:c1:87:c7:8f:dc:5b:04:43:41:97:
38:b9:e7:ca:07:2b:72:f6:d6:f0:d8:ad:3d:69:ec:
50:9f:1a:30:02:50:10:52:99:08:a5:18:1b:69:5e:
1a:4d:27:04:79:d2:b0:5c:d5:f3:56:41:1f:de:26:
79:89

publicExponent: 65537 (0x10001)

privateExponent:

18:da:57:6a:0a:33:b1:e8:42:a2:86:c5:a6:a7:ed:
76:fa:fb:07:86:91:81:c9:8c:1c:42:52:52:57:ff:
fd:e8:ed:89:78:18:75:e9:f7:85:32:03:97:8b:62:
48:50:c8:3c:b3:39:b8:70:85:1c:63:7e:c7:d0:e4:
7b:c6:b9:a5:a6:30:19:a4:e0:3e:86:a1:40:d9:e2:
ea:db:90:5f:47:d6:ac:0e:c3:b3:e8:e9:55:b5:b9:
f2:21:c3:f8:da:89:81:a0:8e:50:46:2b:a3:eb:c2:
18:b6:e2:25:19:ea:13:2f:90:77:ed:ed:6a:c5:12:
28:18:6c:10:50:c4:66:fe:1b:77:c9:a8:76:55:15:

:
█

```
publicExponent: 65537 (0x10001)
privateExponent:
  18:da:57:6a:0a:33:b1:e8:42:a2:86:c5:a6:a7:ed:
  76:fa:fb:07:86:91:81:c9:8c:1c:42:52:52:57:ff:
  fd:e8:ed:89:78:18:75:e9:f7:85:32:03:97:8b:62:
  48:50:c8:3c:b3:39:b8:70:85:1c:63:7e:c7:d0:e4:
  7b:c6:b9:a5:a6:30:19:a4:e0:3e:86:a1:40:d9:e2:
  ea:db:90:5f:47:d6:ac:0e:c3:b3:e8:e9:55:b5:b9:
  f2:21:c3:f8:da:89:81:a0:8e:50:46:2b:a3:eb:c2:
  18:b6:e2:25:19:ea:13:2f:90:77:ed:ed:6a:c5:12:
  28:18:6c:10:50:c4:66:fe:1b:77:c9:a8:76:55:15:
  96:2b:f0:ff:1e:bf:99:aa:4a:ac:13:21:12:2b:76:
  6f:7a:10:e1:37:af:34:aa:e4:54:ea:9b:49:ba:1e:
  f4:89:53:cc:7a:69:86:86:ee:6c:aa:f6:a4:0c:ec:
  67:d2:1d:df:59:df:0f:1a:7f:e8:b1:94:29:8d:6b:
  fc:d5:31:1d:86:d8:ce:f5:fc:b1:e7:89:b4:59:ed:
  d2:da:50:31:4b:1b:9c:10:f7:f4:fb:52:4c:d7:ba:
  5a:6e:c8:c2:ef:45:a3:d1:86:f7:f5:45:6e:fb:43:
  3b:b3:08:05:80:cf:92:36:18:9c:08:0f:96:a0:d3:
  c7
prime1:
  00:e2:7f:af:13:85:b0:1c:99:b2:76:79:96:8c:2d:
  81:5d:53:c6:79:af:49:07:10:9f:1d:bd:ea:d3:cc:
  35:b8:e6:bd:bb:5a:11:97:79:37:11:05:f0:cc:92:
  27:49:35:94:50:2d:69:5f:7c:c3:ab:ea:2a:da:c9:
  62:8a:ec:51:b8:b1:bc:93:e4:69:c4:8b:58:08:b1:
  39:49:fb:4f:f0:8c:3e:40:0d:e0:a3:9a:39:75:3a:
  ee:5b:0e:8b:9e:78:fa:29:01:74:b6:da:ae:3c:11:
  e5:21:7e:4d:1e:c7:03:eb:7b:f8:e7:9f:8a:8a:e9:
  06:9a:b8:be:af:0d:8e:37:ef
prime2:
:
```


prime2:

00:b5:b4:2a:6a:ef:8e:13:72:87:c9:4a:6b:41:23:
bd:50:26:06:37:08:06:70:98:63:bc:f2:16:09:9d:
8e:69:76:c4:2e:97:7d:8b:6a:d6:24:1a:b9:d6:17:
b8:ba:9d:ab:5f:96:b2:d6:4e:b3:b9:88:cc:b1:64:
1c:bc:46:d9:50:63:1c:f5:ad:0f:0d:26:6f:9d:8a:
2b:ad:05:7a:e6:3f:dc:a3:18:7f:b3:87:cf:56:c2:
ad:a5:c5:cd:b3:19:c7:21:a9:44:1c:9e:6d:72:7d:
57:50:91:a1:a6:ce:53:67:dc:26:fb:4b:d6:d8:0b:
4c:63:56:94:26:7c:2a:2e:07

exponent1:

5c:35:0e:d9:2b:bb:9e:69:13:b6:17:99:aa:47:7c:
3a:ba:38:3a:a1:99:61:3e:eb:07:f8:a0:f8:c8:51:
b6:8c:79:01:24:91:b4:d0:ac:12:4a:10:d5:b7:4a:
37:0b:90:ed:73:0f:4a:c8:48:00:5c:bc:45:85:66:
90:97:d4:4d:14:4e:69:06:bb:e8:3f:b1:ff:17:e6:
f2:af:54:c3:e0:d4:16:cf:8f:7c:0a:f5:09:3c:8c:
7c:95:57:51:94:7e:82:e3:0c:d8:c3:fd:da:86:8e:
79:2c:29:c9:35:cb:ea:b9:f5:82:95:f8:98:c4:74:
e3:d8:02:08:d1:c3:ec:11

exponent2:

34:b5:6d:b4:95:87:2e:47:03:f2:b5:3e:c5:a7:9b:
6c:1b:8b:a3:fd:14:f9:26:8b:90:2c:dd:8d:ae:81:
b8:40:36:8a:99:71:56:b7:08:25:73:07:43:7e:a0:
1c:aa:4e:23:2d:70:f5:4f:e1:3c:c9:dd:2e:fc:2e:
85:c3:3b:98:e1:17:6e:80:e4:23:d9:69:79:56:b2:
72:bf:6b:36:4d:82:30:61:93:f1:bd:4c:bd:0b:3e:
1a:1f:76:1b:25:38:29:a0:93:5c:85:ec:93:d0:9e:
3a:85:f4:51:91:fa:89:df:48:03:8f:ad:64:00:6b:

:

```

coefficient:
  00:a0:70:c7:a9:16:12:eb:1e:02:15:36:11:86:76:
  3a:f0:06:88:9c:34:07:97:6b:a6:4a:69:50:af:66:
  74:d7:ce:bb:ae:78:08:6e:38:9f:a4:7e:13:91:b5:
  51:52:2a:af:b0:56:bd:37:0b:54:2a:6c:3d:5d:97:
  65:69:6f:4b:6e:e7:c9:b7:eb:bb:8e:a1:82:bf:64:
  88:2a:bf:04:77:aa:55:b4:1e:35:fc:bf:74:db:e6:
  e1:6f:cf:7a:8d:ad:8b:97:9d:43:73:69:35:aa:c7:
  e2:2b:dc:ce:a6:2c:55:de:c9:4a:42:f0:0f:9f:d7:
  b9:27:23:e2:00:f2:eb:e0:41
-----BEGIN PRIVATE KEY-----
MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBCwggSjAgEAAoIBAQCgw64XdsoejCx9
t5zI0ThIEyc3Nm09n3/9IWF20SOJaQDjwQfyarD6sdQzKqKKmNQeg3DL8hVN9VU
prlbDrqphFvRlSPTwMZJvBmKCYGoTRC8vFbwIn8jPMXXvcAqAgFX5Tb3NNh1KCNj
AtbM8rN0NbXggEsDIgVpV9jneTCFR/bT2RiUCGc9K/1dFctbX9zZcg8CmLIqhVvD
Ifaz0NGRHYFLB6SH6EZKx9HZenZ0aokrgelPaUML18ikjC14NjzyX8K0fCP4wYfH
j9xbBENBlzi558oHK3L21vDYrT1p7FCfGjACUBBSmQilGBtpXhpNJwR50rBc1fNW
QR/eJnmJAgMBAAECggEAGNpXagozsehCoobFpqftdvr7B4aRgcmMHEJSUlf//ejt
iXgYden3hTIDl4tiSFDIPLM5uHCFHGN+x9Dke8a5paYwGaTgPoahQNni6tuQX0fw
rA7Ds+jpVbW58iHD+NqJgaCOUEYro+vCGLbiJRnqEy+Qd+3tasUSKBhsEFDEZv4b
d8modlUVliw/x6/mapKrBMhEit2b3oQ4TevNKrkV0qbSboe9ILTzHpphobubKr2
pAzsZ9Id31nfDxp/6LGUKY1r/NUxHYbYzvX8seeJtFnt0tpQMUsbnBD39PtSTNe6
Wm7Iwu9Fo9GG9/VFbvtD07MIBYDPkYjYnAgPlqDTxwKBgQDif68ThbAcmbJ2eZaM
LYFdU8Z5r0kHEJ8dverTzDW45r27WhGXeTcRBfDMkidJNZRQLWlffM0r6irayWKK
7FG4sbyT5GnEi1gIsTLJ+0/wjD5ADeCjnmjl10u5bDoueePopAXS22q48EeUhfK0e
xwPre/jnn4qK6QaaUL6vDY437wKBgQC1tCpQ744TcofJSmtBI71QJgY3CAZwmGO8
8hYJnY5pdsQul32LatYkGrnWF7i6natflrLWTr05iMyxZBy8RtlQYxz1rQ8NJm+d
iutBXrmP9yjGH+zh89Wwq2lxc2zGcchqUQcnm1yfVdQkaGmzlNn3Cb7S9bYC0xj
VpQmfCouBwKBgFw1Dtkru55pE7YXmapHfDq60DqhmWE+6wf4oPjIUbaMeQEkkbTQ
rBJKENW3SjcLk01zD0rISABcvEWFZpCX1E0UTmkGu+g/sf8X5vKvVMPg1BbPj3wK
9Qk8jHyVV1GUfoLjDNjD/dqGjnksKck1y+q59YKV+JjEd0PYAgjRw+wRAoGANLVt
:

```

- i. Which values are constant?

The values that stayed constant in both cases are the key size (2048 bits) and the public exponent (65537 or 0x10001), which are common RSA settings and are often chosen as the standard parameters.

- ii. Which ones vary?

The values that were basically changed between the 2 keys were the modulus, private exponent, prime 1, prime 2, exponent 1, exponent 2, and the coefficient. These values differ for both keys because each RSA key

pair is generated using new prime numbers each time, which in turn makes the mathematical values unique to each key.

- iii. What do these values represent?

These values are basically the mathematical parts of an RSA key pair. Over here, the modulus basically represents the product of the 2 prime numbers, and it is a part of both the public and private key. The public exponent is the number used for encryption or signature verification. The private exponent is the number used for decryption or signing. Prime1 and Prime2 basically represent the 2 secret prime numbers that are used to build the modulus. The Exponent 1 and 2 values are basically used in the Chinese Remainder Theorem optimization, which is used to make decryption or signing faster and Coefficient is basically another CRT value which could be used to speed up private key operations.

(15 points, 5 points each question)

3. **Discussion of the differences between FTP and SFTP.**

- i. Why would you want one over the other?

The major difference between FTP and SFTP is that FTP is not secured that is it sends data ,usernames and passwords in plaintext, which could be intercepted and in the lab we basically use it to upload public key since security was not a critical aspect for that step, whereas SFTP is secured since it uses SSH encryption to protect any data during transfer and in the lab we used to download messages securely since it protects the data from getting intercepted and as compared to FTP it also helps in preventing password theft, eavesdropping and unauthorized access.

- ii. Why did we need to specify our private key?

In our lab we used the command, `sftp -i pulkit_private_key.pem ssh_guest@certs.homework.3310.com` to specify our private key and it was important for us to do that because SFTP uses public or private key authentication instead of passwords and over here, the server already has our public key which we uploaded via FTP, so the server needs to verify that we are the correct user and can basically login without even needing a password and that's why we provide our private key to just basically prove our identity.

- iii. What protection does this offer?

The public/private key authentication basically provides strong security benefits such as authentication (it only allows the person having the correct private key to access the

server), confidentiality(SFTP basically encrypts all data during transfer) and integrity(the data cannot be modified during transmission without any detection).

(15 points, 5 points each question)

4. **Screenshot of the five messages [netid]1.txt, [netid]2.txt, ... [netid]5.txt**

(10 points)

```
cpre3310@cpre3310:~/homework/lab08/pulkit$ openssl dgst -sha256 -verify lab08_public_key.pem -signature sig.txt.sha256 p
ulkit1.txt 2>/dev/null
Verification failure
cpre3310@cpre3310:~/homework/lab08/pulkit$ openssl dgst -sha256 -verify lab08_public_key.pem -signature sig.txt.sha256 p
ulkit2.txt 2>/dev/null
Verification failure
cpre3310@cpre3310:~/homework/lab08/pulkit$ openssl dgst -sha256 -verify lab08_public_key.pem -signature sig.txt.sha256 p
ulkit3.txt 2>/dev/null
Verification failure
cpre3310@cpre3310:~/homework/lab08/pulkit$ openssl dgst -sha256 -verify lab08_public_key.pem -signature sig.txt.sha256 p
ulkit4.txt 2>/dev/null
Verified OK
cpre3310@cpre3310:~/homework/lab08/pulkit$ openssl dgst -sha256 -verify lab08_public_key.pem -signature sig.txt.sha256 p
ulkit5.txt 2>/dev/null
Verification failure
cpre3310@cpre3310:~/homework/lab08/pulkit$
```

5. **Discussion on hash verification**

i. What is known about the message?

By looking at our verification results we can say that only pulkit4.txt is the authentic message because it returned “Verified OK”, which means that the hash of this message matches the provided digital signature when it is verified using the public key and that’s why the message has not been signed and it can be trusted as the original message, whereas the other messages resulted in “Verification failure”, which means that their contents were altered and their hashes don’t match the signature now.

ii. What is the message protected against and what is it vulnerable to?

The message is protected against any kind of modification, since if any part of the message is changed, its hash will change as well which causes the verification to fail too. This basically ensures data integrity and confirms that the message has not been altered. Also, since the signature is verified using a public key, it also provides authentication, which means that the message was signed by the holder of the corresponding private key, however, since the message is not encrypted

(10 points, 5 points each question)

6. **Discussion on what the message generated in step 8e protected against and what it is vulnerable to (compared to the message we downloaded in step 6).**

(10 points, 5 points each part)

The encrypted message is basically protected against any kind of unauthorized authentication and since the message was encrypted using the lab08 public key, we can say that only the holder of the corresponding private key can decrypt the message and read it, which ensures that even if intercepts the file (which is pulkit_encrypt.dat), they cannot understand its contents since it is encrypted, unlike the messages that were downloaded earlier which were not encrypted, so anyone could read them.

The message is basically vulnerable to any any kind of modification, since the message is only encrypted and not digitally signed, so an attacker could modify and replace the encrypted file and there is no way to verify who sent the message. Additionally, the receiver cannot confirm whether the the message was altered.

7. **Screenshot of the signed certificate ([netid]_certificate.pem) when looked at through openssl**
(10 points)

```
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number: 12 (0xc)
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C = US, ST = Iowa, L = Ames, O = 3310.com, OU = homework, CN = certs.homework.3310.com, emailAddress = c
    erts@homework.3310.com
    Validity
      Not Before: Mar 27 00:30:01 2026 GMT
      Not After : Mar 27 00:30:01 2027 GMT
    Subject: C = US, ST = Iowa, O = 3310.com, OU = homework, CN = pulkit.homework.3310.com, emailAddress = pulkit@ho
    mework.3310.com
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (2048 bit)
      Modulus:
        00:a2:44:02:81:0a:ed:ae:8d:e0:d0:6a:3a:63:4f:
        dd:dd:45:04:8b:ee:bf:8d:74:6d:80:a5:75:a1:13:
        da:c4:d9:9f:89:66:25:e8:fa:9c:ea:39:17:06:84:
        3b:39:1d:41:21:b7:7c:ee:1b:2f:2c:35:d4:43:29:
        71:af:d7:37:58:56:79:de:25:fa:d5:39:14:43:77:
        49:df:05:00:fe:90:a1:99:40:cb:d3:89:a3:be:94:
        28:22:24:77:07:1a:1d:0d:9f:0a:42:6d:d7:b0:fd:
        d0:68:12:7b:e1:55:28:76:0d:71:74:dc:ec:62:54:
        ac:ec:fa:92:76:ca:88:c4:f0:5a:13:52:89:54:3f:
        30:3a:bd:94:a3:c7:da:d0:8f:ca:73:2e:4e:6b:7d:
        ef:cf:ae:31:56:6a:cc:9e:d5:0b:36:87:f9:ab:b8:
        35:cf:50:4c:c0:b0:13:86:0f:fa:7f:9d:3a:b3:7b:
        9b:63:dc:54:4a:45:6e:92:a0:ce:e1:47:59:eb:20:
        51:e0:79:2d:ec:1b:1a:d8:7f:ab:42:40:73:8e:76:
        36:a9:d3:79:e7:61:67:5b:47:b3:02:4f:25:f9:d7:
```

```
cpre3310@cpre3310: ~/homework/lab08
36:a9:d3:79:e7:61:67:5b:47:b3:02:4f:25:f9:d7:
38:33:41:31:00:16:0b:c6:5c:44:6a:24:93:bb:db:
c2:23:de:3a:6a:e5:62:41:68:3a:39:19:45:54:c5:
71:09
Exponent: 65537 (0x10001)
X509v3 extensions:
  X509v3 Basic Constraints:
    CA:FALSE
  Netscape Comment:
    OpenSSL Generated Certificate
  X509v3 Subject Key Identifier:
    3A:27:DF:E6:EE:F0:F0:EB:0E:B2:38:C0:84:87:91:87:6E:D5:15:8C
  X509v3 Authority Key Identifier:
    5A:47:BD:66:5F:78:80:7C:00:09:CE:12:AF:9F:AE:23:F2:8E:65:78
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
  95:a0:bb:f5:01:a0:61:bd:a9:a0:cf:90:f0:70:90:bd:c9:74:
  32:37:db:7a:70:38:5d:e9:13:86:bf:ae:a3:6e:be:e4:e9:d5:
  83:81:cd:56:ae:4d:dc:7a:2e:e0:b8:30:4f:bc:19:ae:0a:54:
  9c:98:3d:6e:3e:4c:78:f4:98:6f:72:63:6e:ce:6b:83:70:0e:
  b2:2c:a4:b4:6b:47:5d:f3:41:13:d3:d1:c9:83:88:7a:42:4d:
  1d:5a:e8:2f:74:9a:82:ad:6f:82:68:08:b1:0c:aa:8d:ea:40:
  72:a3:35:1c:c0:8e:4d:62:29:b5:2d:75:f3:83:f1:b4:bf:1f:
  35:99:67:a7:43:17:4f:44:63:de:b4:83:8e:9b:69:5d:e5:e9:
  b9:64:8e:39:3f:f0:3a:6e:3e:f7:76:8a:b9:01:8f:c0:93:74:
  c9:83:bb:81:0b:f8:61:56:58:45:f3:09:97:85:9e:a9:03:d3:
  01:e5:0c:e4:1c:25:04:af:ac:1d:bf:d7:4a:30:c2:e4:70:2e:
  9c:d3:7c:58:5e:2b:e2:62:d7:21:ca:32:1d:85:1a:9a:e0:bf:
  1b:1f:60:6e:af:4d:26:03:46:69:b4:3d:2e:c8:93:61:60:59:
  71:25:ac:2b:63:c0:99:1e:ae:31:54:5d:42:57:68:ce:54:bf:
  9f:d0:fc:e8
```

8. Discussion from step 12

- i. Do any parts of the certificate match with your private key? If so, why?

Yes, some parts of my certificate match with my private key, especially the public key inside the certificate corresponds to my private key. This happens because the certificate was generated using my private key and the corresponding public key is basically embedded in the certificate. Since a public key or private key pair is mathematically linked to each other, the public key in the certificate matches my private key.

- ii. What was happening during the Certificate Signing process? Why did you need to submit it for signing?

During the certificate signing process, I basically created a Certificate Signing Request (CSR). This CSR basically contains my public key and my identity information (like name, information and domain) and basically when I uploaded my the CSR, the Certificate Authority (CA) verified the request and then basically signed it using its own private key, which creates a valid certificate. This process is very important because it confirms that the public key belongs to me and it allows others to basically trust my certificate using the CA's public key.

I needed to submit the CSR for signing so that my certificate would be trusted by others, without the CA's signature the certificate may not be considered valid.

(20 points, 10 points each answer, 10 points each why)